

Drive for better vision



WE2 SDK User Guide V1.2

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Agenda

- **Environment Setting**

- ❖ PC Environment Request
- ❖ Setting GNU

- **Firmware Image Generation**

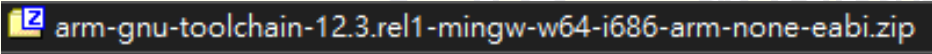
- ❖ Setting makefile
- ❖ Compiling
- ❖ Generating firmware image

- **How to Burn Firmware Image to EVB**

- ❖ Burn by PC Tool
- ❖ Upgrade through OTA Tool

Environment Setting

Environment Setting

- PC environment request (minimum-hardware-requirements)
 - ❖ HW: x86 64bit
 - ❖ OS: Win10 64bit
- Setting GNU by following steps
 - ① Download GNU
by following link (arm-gnu-toolchain-12.3.rel1-mingw-w64-i686-arm-none-eabi.zip)
(<https://developer.arm.com/-/media/Files/downloads/gnu/12.3.rel1/binrel/arm-gnu-toolchain-12.3.rel1-mingw-w64-i686-arm-none-eabi.zip>)

 - ② Unzip the .zip to SDK_root
 - ③ **[Must]** Check 1. if the folder exists: “SDK_root/arm-gnu-toolchain-12.3.rel1-mingw-w64-i686-arm-none-eabi/”
and 2. its content is like the sample below.

↓ Sample: The content of folder: “SDK_root/arm-gnu-toolchain-12.3.rel1-mingw-w64-i686-arm-none-eabi”

arm-gnu-toolchain-12.3.rel1-mingw-w64-i686-arm-none-eabi			
名稱	修改日期	類型	大小
arm-none-eabi	2023/6/28 上午 09:34	檔案資料夾	
bin	2023/6/29 下午 10:28	檔案資料夾	
include	2023/6/28 上午 06:20	檔案資料夾	
lib	2023/6/28 上午 06:26	檔案資料夾	
libexec	2023/6/28 上午 06:26	檔案資料夾	
share	2023/6/29 下午 10:11	檔案資料夾	
.version	2023/6/28 上午 09:44	VERSION 檔案	1 KB
12.3.rel1-mingw-w64-i686-arm-non-...	2023/6/29 下午 10:11	文字文件	5 KB

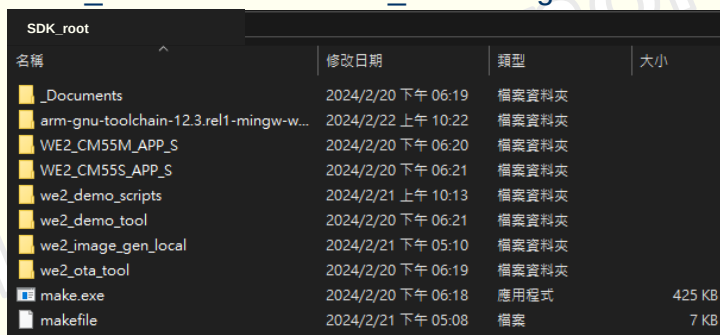
Firmware Image Generation

Firmware Image Generation (1)

- Set the following makefile respectively:

1. **makefile:** (path: `SDK_root/makefile`)

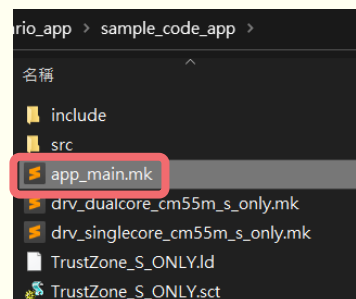
- ① `CORE_SETTING = SINGLE_CORE`
- ② `EVK = HX6538_AIoT_EVB_QFN88_V10`
- ③ `APP_TYPE = sample_code_app`
- ④ `GNU_TOOLPATH = "SDK_root\arm-gnu-toolchain-12.3.rel1-mingw-w64-i686-arm-none-eabi\bin"`



名稱	修改日期	類型	大小
._Documents	2024/2/20 下午 06:19	檔案資料夾	
arm-gnu-toolchain-12.3.rel1-mingw-w...	2024/2/22 上午 10:22	檔案資料夾	
WE2_CM55M_APP_S	2024/2/20 下午 06:20	檔案資料夾	
WE2_CM55S_APP_S	2024/2/20 下午 06:21	檔案資料夾	
we2_demo_scripts	2024/2/21 上午 10:13	檔案資料夾	
we2_demo_tool	2024/2/20 下午 06:21	檔案資料夾	
we2_image_gen_local	2024/2/21 下午 05:10	檔案資料夾	
we2_ota_tool	2024/2/20 下午 06:19	檔案資料夾	
make.exe	2024/2/20 下午 06:18	應用程式	425 KB
makefile	2024/2/21 下午 05:08	檔案	7 KB

2. **app_main.mk** (path: `SDK_root/WE2_CM55M_APP_S/app/scenario_app/<scenario>/app_main.mk`)
(sample: `SDK_root/WE2_CM55M_APP_S/app/scenario_app/sample_code_app/app_main.mk`)

- ① `TFLITE_ALGO = YOLO_FASTEST`
- ② `CIS = HM0360_MONO`
- ③ `TRANSMIT_PROTOCOL = SPI_MASTER`



Firmware Image Generation (2)

- Execute the following operations step-by-step:

① Launch cmd.exe on PC and cd to SDK_root

② Execute command: **make clean**

③ Execute command: **make all**

➢ Make sure there are NO ERROR(s) at output message.

➢ If the compilation is completed, these files should be generated in the path:

path: SDK_root/WE2_CM55M_APP_S/obj_epii_evb_icv30_bdv10/gnu_epii_evb_QFN88/

(file: 1. WE2_CM55M_gnu_epii_evb_QFN88_s.elf

2. WE2_CM55M_gnu_epii_evb_QFN88_s.map)

④ Execute command: **make flash**

➢ Make sure the .elf and .map has been generated at the last step.

➢ If the flash is completed, the output.img should be generated at:

path: SDK_root/we2_ota_tool/img/output.img

⑤ Check the output.img has been generated.

1. Sample: cd to SDK_root

```
C:\Windows\System32\cmd.exe
D:\workSpace\sdk_package\ SDK_root
```

2. Sample: output message when “make clean” is complete.

```
C:\Windows\system32\cmd.exe /C if exist obj_epii_evb_icv30_bdv10\gnu_epii_evb_QFN88
C:\WINDOWS\system32\cmd.exe /C if exist .sc.project C:\
make[1]: Leaving directory 'D:/workSpace/sdk_package/Himax'
clean complete.
```

3. Sample: output message when “make all” is complete.

```
text      data      bss       dec       hex filename
27695      324      1596     29615     73af obj_epii_evb_icv30_bdv10\gnu_epii_evb_QFN88_s.elf
make[1]: Leaving directory 'D:/workSpace/sdk_package/Himax'
build complete.
```

4. Sample: output message when “make flash” is complete

```
IMAGE GEN DONE
copy we2_image_gen_local\output_case3_secboot_nodiv\output
複製了 1 個檔案。
```

5. Sample: the generated output.img.

```
SDK_root > we2_ota_tool > img
名稱
output.img
```

How to Burn Firmware Image to EVB

- The generated firmware can be burned to the EVB by these method:
 - ❖ **Burn by PC Tool:**
Refer to “2_EVK_and_PCTool_User_Guide_HX6538_AIoT_EVB_QFN88” to find the instructions about how to burn the output.img.
 - ❖ **Upgrade through OTA Tool:**
Refer to “3_OTA_User_Guide” to learn about how to use OTA to do FW upgrade.

HINT

- ❖ If there is no firmware in EVK, it can only be burned through PC Tool.
- ❖ Path of image: SDK_root/we2_ota_tool/img/**output.img**